

1. $m(\theta) = E(X) = 1 - \theta^2$

$\mathcal{M} = (-1, 1)$

$\hat{\theta} = \underset{\theta \in \mathcal{M}}{\operatorname{argmin}} Q(\theta)$

$$Q(\theta) = \sum_{i=1}^5 (x_i - m(\theta))^2$$

$$= \sum_{i=1}^5 (x_i - (1 - \theta^2))^2$$

$$= \sum_{i=1}^5 (\theta^2 + x_i - 1)^2$$

$$Q'(\theta) = \sum_{i=1}^5 2\theta \cdot 2(\theta^2 + x_i - 1) = 0$$

$\Leftrightarrow (\theta \neq 0)$

$$\sum (\theta^2 + x_i - 1) = 0 \Leftrightarrow 5\theta^2 - 5 + \sum x_i = 0$$

$$\Leftrightarrow 5\theta^2 = 5 - \sum x_i$$

$$\theta = \pm \sqrt{1 - \bar{X}} = \pm \sqrt{1 - 0.4402}$$

$$= \pm 0.7482$$

Sign study

θ	-0.7482	0	0.7482
$Q'(\theta)$	-	0	+
$Q(\theta)$	$\searrow$ min $\nearrow$		$\searrow$ min $\nearrow$

Thus there are two values of θ that minimize $Q(\theta)$ (they both the same result) and therefore $\hat{\theta} = \pm 0.7482$

2. X number of times A correctly foresees future.
 Y - 11 - B - 11 -

$X \in \text{Bin}(10, \theta)$, $Y \in \text{Bin}(21, \theta)$ with $\theta = P(\text{foresee the future correctly})$.

X, Y independent. $X + Y =: Z \in \text{Bin}(31, \theta)$

a) $Z = X + Y = 4 + 15 = 19$ observation of Z

$$L(\theta) = \binom{31}{19} \theta^{19} (1-\theta)^{31-19} \Rightarrow \hat{\theta} = \underset{\theta}{\text{argmax}} L(\theta) = \frac{19}{31}$$

b) $H_0: \theta \leq 1/2$

$H_1: \theta > 1/2$

p-value method

$$p = \sup_{\theta \leq 1/2} P_{\theta}(Z \geq 19)$$

$$= P_{1/2}(Z \geq 19)$$

is more than 0.05

$$3. \quad f(x) = \begin{cases} \frac{1}{4} \binom{25}{-x} \theta^{-x} (1-\theta)^{25+x} & x = -25, \dots, -1, 0 \\ \frac{3}{4} \theta e^{-\theta(x-1)} & x \geq 1 \end{cases}$$

$\Theta = [0, 1]$ parameter space

$$a) \quad L(\theta) = \left(\frac{1}{4}\right)^4 \prod_{x \in \{-11, -15, -12, -11\}} \binom{25}{-x} \theta^{-x} (1-\theta)^{25+x} \cdot \left(\frac{3}{4}\right)^5 \prod_{i=5}^9 \theta e^{-\theta(x_i-1)}$$

$$\begin{aligned} l(\theta) &\sim \log \theta \cdot \underbrace{(11+15+12+11)}_{49} + \log(1-\theta) \cdot \underbrace{(100-49)}_{51} \\ &\quad + 5 \log \theta - \theta (2.42 + 3.03 + 2.65 + 1.88 + 1.03 - 5) \\ &= 54 \log \theta + 51 \cdot \log(1-\theta) - 6.01 \cdot \theta \end{aligned}$$

$$l'(\theta) = \frac{54}{\theta} - \frac{51}{1-\theta} - 6.01 = 0 \quad \Leftrightarrow$$

$$(-54 - 51 - 6.01)\theta + 6.01\theta^2 + 54 = 0 \quad \Leftrightarrow$$

$$(\theta - 9.2355)^2 = 76.3045 \quad (=)$$

$$\theta_1 = 0.5$$

$$(\theta_2 = 17.9710) \notin \Theta$$

3)

Sign Study

3 (contd)

θ	0.5	
$V'(\theta)$	+	0
$V(\theta)$	$\rightarrow$	max

Thus $\theta = 0.5$ maximum

$$3b) \quad \eta = h(\theta) = P_{\theta}(-3 \leq \bar{X} \leq 3 \mid \bar{X} \geq -1)$$

$$= \frac{P(-3 \leq \bar{X} \leq 3 \cap \bar{X} \geq -1)}{P(\bar{X} \geq -1)}$$

$$= \frac{P(-1 \leq \bar{X} \leq 3)}{P(\bar{X} \geq -1)}$$

$$= \frac{P(\bar{X} = -1) + P(\bar{X} = 0) + \int_1^3 f(u) du}{P(\bar{X} = -1) + P(\bar{X} = 0) + \int_1^{\infty} f(u) du}$$

$$= \frac{\frac{1}{4} \left[\binom{25}{1} e(1-e)^{24} + \binom{25}{0} (1-e)^{25} \right] + \frac{3}{4} (1-e^{-2e})}{\frac{1}{4} \left[\binom{25}{1} e(1-e)^{24} + \binom{25}{0} (1-e)^{25} \right] + \frac{3}{4}}$$

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$$\hat{\eta} = h(\hat{\theta}) = \frac{\frac{1}{4} [25 \cdot 0.5^{25} + 1 \cdot 0.5^{25}] + \frac{3}{4} (1-e^{-1})}{\frac{1}{4} [25 \cdot 0.5^{25} + 1 \cdot 0.5^{25}] + \frac{3}{4}}$$

$$= \dots \approx 0.6321$$

$$4. \quad \bar{X}_1 \sim \text{Po}\left(\frac{1}{\theta}\right), \quad \bar{X}_2 \sim \text{Po}\left(\frac{2}{\theta}\right)$$

$\bar{X}_1, \bar{X}_2$ independent.

$$x_1 = 7, x_2 = 23$$

$$(u) = [0, \infty)$$

$$a) \quad \text{MLE } \hat{\theta} = \underset{\theta \in (u)}{\text{argmin}} \sum_{i=1}^2 (x_i - m(\theta))^2 = \underset{\theta \in (u)}{\text{argmin}} Q(\theta)$$

$$Q(\theta) = \left(7 - \frac{1}{\theta}\right)^2 + \left(23 - \frac{2}{\theta}\right)^2$$

$$Q'(\theta) = -2\left(7 - \frac{1}{\theta}\right)\frac{1}{\theta^2} - 2\left(23 - \frac{2}{\theta}\right)\frac{2}{\theta^2} = 0$$

$$\Leftrightarrow (\theta \neq 0)$$

$$-2(7\theta - 1) - 2(23\theta - 2)2 = 0$$

$\Leftrightarrow$

$$\theta = \frac{5}{53}$$

$$\left(\frac{5}{x_1 + 2x_2} \right)$$

$$b) \quad Z = \bar{X}_1 + \bar{X}_2 \sim \text{Po}\left(\frac{1}{\theta} + \frac{2}{\theta}\right) = \text{Po}\left(\frac{3}{\theta}\right)$$

large values of Z point to small values of θ

p-value method

$$p = \sup_{\theta \geq 1/5} P_{\theta}(Z \geq 30)$$

$$H_0: \theta \geq 1/5$$

$$H_1: \theta < 1/5$$

$$= P_{1/5}(Z \geq 3) = 1 - P_{1/5}(Z \leq 25)$$

$$7+23$$

$$= (\text{table 5}) = 1 - 1 = \underline{0}$$

5)

4c) Test (T, C) given $T = X_1 + X_2$
 $C = [35, \infty)$

$$\bar{\alpha}(\theta) = P_{\theta}(T \in C) = P_{\theta}(\underbrace{\bar{X}_1 + \bar{X}_2}_{P_0(\frac{3}{\theta})} \geq 35)$$

When $\theta = \frac{1}{15}$, $\frac{3}{\theta} = 3 \cdot 15 = 45$ large enough
 for Gaussian approximation, so $P_0(\eta) \approx N(\eta, \eta)$
↑
variance

Thus

$$\bar{\alpha}(\frac{1}{15}) = P_{1/15}(\frac{\bar{X}_1 + \bar{X}_2 - 45}{\sqrt{45}} \geq \frac{35 - 45}{\sqrt{45}})$$

$$\stackrel{\substack{\nearrow \\ U \sim N(0,1)}}{\approx} P(U \geq \frac{-10}{\sqrt{45}}) = \cancel{P(U \geq \dots)} \text{ (symmetry)}$$

$$= P(U \leq \frac{10}{\sqrt{45}}) = \Phi(1.4907) \approx 0.9319$$

↑
 Φ

$$5 \quad x_1, \dots, x_4 \sim N(\mu_1, \sigma_A^2)$$

$$y_1, \dots, y_5 \sim N(\mu_2, \sigma_B^2)$$

$$a) \quad H_0: \sigma_A^2 \leq \sigma_B^2$$

$$H_1: \sigma_A^2 > \sigma_B^2$$

$$T = \frac{\frac{s_A^2}{\sigma_A^2}}{\frac{s_B^2}{\sigma_B^2}} \in F(4-1, 5-1) = F(3, 4)$$

A test of H_0 then becomes (sup over all σ_A^2, σ_B^2 in H_0 it attained for $\sigma_A^2 = \sigma_B^2$): reject H_0 if

$$T = \frac{s_A^2}{s_B^2} > F_{0.05}(3, 4) = 6.59 \quad (\text{from table 7})$$

We have

$$\begin{aligned} s_A^2 &= \frac{1}{4-1} \sum_{i=1}^4 (x_i - \bar{x})^2 \\ &= \frac{1}{3} (\sum x_i^2 - 4(\bar{x})^2) = 3.7740 \end{aligned}$$

$$\begin{aligned} s_B^2 &= \frac{1}{5-1} (\sum y_i^2 - 5(\bar{y})^2) \\ &= \frac{1}{4} (\sum y_i^2 - 5(\bar{y})^2) = 1.8663 \end{aligned}$$

$\Rightarrow$

$$T = 2.0222 < F_{0.05}(3, 4) \Rightarrow \text{do not reject } H_0$$

(8)

5 b) If $\sigma_A^2 = \sigma_B^2 =: \sigma^2$ then

$$s_p^2 = \frac{3s_A^2 + 4s_B^2}{3+4}$$

and

$$\frac{7s_p^2}{\sigma^2} \in \chi^2(7)$$

Thus

$$0.99 = P(\chi_{0.99}^2(7) < \frac{7s_p^2}{\sigma^2})$$

$$= P\left(\frac{\chi_{0.99}^2(7)}{7s_p^2} < \frac{1}{\sigma^2}\right)$$

$$= P\left(\sigma^2 < \frac{7s_p^2}{\chi_{0.99}^2(7)}\right)$$

$$= P\left(-\sigma^2 > -\frac{7s_p^2}{\chi_{0.99}^2(7)}\right)$$

$$= P\left(e^{-\sigma^2} > e^{-(7s_p^2/\chi_{0.99}^2(7))}\right)$$

$\Rightarrow$

$$\underline{I} = \left(e^{-7s_p^2/\chi_{0.99}^2(7)}, \infty\right)$$

$$= \left(e^{-18.7872/1.24}, \infty\right)$$

$$= \left(2.63 \cdot 10^{-7}, \infty\right)$$

(9)

$$6 \ a) \quad H_0: S(3) \leq \frac{1}{10}$$

$$H_1: S(3) > \frac{1}{10}$$

$$S(t) = 1 - F(t)$$

$$Z := n S_n(t) = n(1 - F_n(t)) \in \text{Bin}(20, S(t))$$

If $\theta = S(3)$ we have

$Z \sim \text{Bin}(20, \theta)$ and want to test

$$H_0: \theta \leq \frac{1}{10}$$

$$H_1: \theta > \frac{1}{10}$$

Z = observed number of values greater than 3
and $\underline{Z = 3}$.

p-value method

$$p = P_{\frac{1}{10}}(Z \geq 3) = 1 - P_{\frac{1}{10}}(Z \leq 2)$$

$$= 1 - 0.6769 \quad \left(\text{using the pmf of a } \text{Bin}(20, \frac{1}{10}) \text{ r.v.} \right)$$

$$= \underline{0.3231}$$

Do not reject H_0 .

$$6b) \quad Z \in \text{Bin}(20, \theta) \approx N(20\theta, 20\theta(1-\theta))$$

$$T = \frac{T - 20 \cdot \frac{1}{10}}{20 \cdot \frac{1}{10} (1 - \frac{1}{10})} \approx N(0, 1) \quad \text{if } \theta = \frac{1}{10}$$

$$= \frac{Z - 2}{1.3416}$$

Test becomes reject H_0 if

$$T = \frac{Z - 2}{1.3416} > \lambda_{0.05} = 1.6449$$

Power function for the test is at $\theta = \frac{1}{5}$

$$\beta\left(\frac{1}{5}\right) = P_{1/5} \left(\frac{Z - 2}{1.3416} > \lambda_{0.05} \right)$$

$$= P_{1/5} \left(\frac{Z - 2}{\sqrt{20 \cdot \frac{1}{5} \cdot \frac{4}{5}}} > \lambda_{0.05} \frac{1.3416}{\sqrt{20 \cdot \frac{1}{5} \cdot \frac{4}{5}}} \right)$$

$$= P_{1/5} \left(\frac{Z - 20 \cdot \frac{1}{5}}{\sqrt{20 \cdot \frac{1}{5} \cdot \frac{4}{5}}} > \lambda_{0.05} \frac{1.3416}{\sqrt{20 \cdot \frac{1}{5} \cdot \frac{4}{5}}} - \frac{20 \cdot \frac{1}{5}}{\sqrt{20 \cdot \frac{1}{5} \cdot \frac{4}{5}}} \right)$$

$$\approx 1 - \Phi(1.2336 - 2.2361)$$

$$= 1 - \Phi(-1.0025) = \Phi(1.0025) \approx \underline{\underline{0.85}}$$